Distance-Unbalancedness of Unicyclic Graphs with Prescribed Cycle Length*

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Abstract

For a connected graph G, let $n_G(u,v)$ denote the number of vertices that are closer to u than to v, and let

$$\mathrm{uB}(G) = \sum_{\{u,v\}\in\binom{V(G)}{2}} |n_G(u,v) - n_G(v,u)|.$$

We study this invariant on unicyclic graphs with a prescribed cycle length. First we give a seven-vertex counterexample to the lower formula asserted in Theorem 1.2 of Su and Tang, *AIMS Mathematics* 9 (2024), 16863–16875: a triangle with a length-two pendant path ending in two leaves has distance-unbalancedness 34, which is smaller than $(7+2)(7-3) = 36$. We then record two exact facts that are independent of the prescribed cycle length: among all unicyclic graphs of order n, the minimum is 0, uniquely attained by C_n, and for fixed cycle length c the endpoint cases $n = c$ and $n = c+1$ have explicit values. For every fixed $c \geq 3$ we prove the asymptotically sharp maximum formula

$$\max\{\mathrm{uB}(G) : G \in \mathcal{U}_{n,c}\} = \frac{1}{2}n^3 + o(n^3),$$

where $\mathcal{U}_{n,c}$ is the class of n-vertex unicyclic graphs whose unique cycle has length c. Finally we give an exhaustive rooted-tree enumeration for small values of $n-c$, explicit low-unbalanced test families, and resulting conjectures for the minimum in $\mathcal{U}_{n,c}$. The finite count used for the bouquet test families is written out explicitly, because this is the easiest place for hidden counting errors to enter.

Keywords. distance-unbalancedness; unicyclic graph; fixed cycle length; extremal graph; rooted tree enumeration.
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1 Introduction

For a connected graph G and two distinct vertices $u, v \in V(G)$, let

$$n_G(u,v) = |\{w \in V(G) : \mathrm{dist}_G(w,u) < \mathrm{dist}_G(w,v)\}|.$$

The distance-unbalancedness of G is

$$\mathrm{uB}(G) = \sum_{\{u,v\}\in\binom{V(G)}{2}} |n_G(u,v) - n_G(v,u)|.$$

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This invariant was introduced by Miklavič and Šparl [1]. It can be viewed as the complete-pair analogue of the Mostar index [5], where the summation is taken over all unordered vertex pairs rather than only over edges. Distance-unbalancedness has since been studied for several graph classes, including trees [2, 3] and merged graphs built from a triangle and a tree [4].

Let $\mathcal{U}_{n,c}$ denote the class of all connected unicyclic graphs with n vertices and unique cycle length c. For a unicyclic graph this unique cycle length is the girth. We study

$$m_c(n) = \min\{\mathrm{uB}(G) : G \in \mathcal{U}_{n,c}\}, \qquad M_c(n) = \max\{\mathrm{uB}(G) : G \in \mathcal{U}_{n,c}\}.$$

The case $c = 3$ is closely related to the formula asserted by Su and Tang [4, Theorem 1.2] for the merged graph $C_3 \cdot T$, obtained by identifying one vertex of a triangle C_3 with one vertex of a tree T. We show that the stated lower formula fails at $n = 7$; the counterexample is small and is displayed in Figure 1. We also give general results and computational evidence for arbitrary fixed c.

The main rigorous conclusions are as follows.

- A seven-vertex graph in $\mathcal{U}_{7,3}$ has uB $= 34$, contradicting the asserted value $(n+2)(n-3) = 36$ at $n = 7$ in the merged-graph class $C_3 \cdot T$.
- Among all unicyclic graphs of order n, the minimum of uB is 0, and the unique extremal graph is C_n.
- For fixed cycle length c, the values at $n = c$ and $n = c + 1$ are explicit.
- For every fixed $c \geq 3$,
$$M_c(n) = \frac{1}{2}n^3 + o(n^3).$$

The exact finite-order minimum for fixed c appears subtler. We therefore separate theorems from computationally supported conjectures.

2 Preliminaries and the rooted-cycle model

Throughout the paper all graphs are finite, simple, and connected. For a positive integer r, write $[r] = \{0, 1, \ldots, r-1\}$. A unicyclic graph with unique cycle C_c can be decomposed uniquely as a cycle with rooted trees attached to the cycle vertices.

Definition 2.1 (Rooted-cycle representation)**.** *Let $C_c = v_0 v_1 \cdots v_{c-1} v_0$. For rooted trees $T_0, \ldots, T_{c-1}$ whose roots are respectively identified with $v_0, \ldots, v_{c-1}$, write*

$$C_c(T_0, T_1, \ldots, T_{c-1})$$

for the resulting unicyclic graph. The root vertex v_i belongs to T_i.

If $x \in V(T_i)$ and $y \in V(T_j)$ with $i \neq j$, then

$$\mathrm{dist}_G(x, y) = \mathrm{dist}_{T_i}(x, v_i) + \mathrm{dist}_{C_c}(v_i, v_j) + \mathrm{dist}_{T_j}(v_j, y). \tag{1}$$

This elementary formula is the basis of the enumeration described in Section 7.

For later counting on a labelled cycle, put $[c] = \{0, 1, \ldots, c-1\}$ and use the notation

$$\rho_c(i, j) = \min\{|i - j|, c - |i - j|\}, \qquad i, j \in [c],$$

and

$$\varepsilon(a,b) = \begin{cases} 1, & a < b, \\ -1, & a > b, \\ 0, & a = b. \end{cases}$$

Thus $\varepsilon(a,b)$ records the signed contribution of a vertex whose distance to the first member of a pair is a and whose distance to the second member is b.

3 A correction in the girth-three case

Su and Tang [4, Theorem 1.2] studied the graph $H = C_3 \cdot T$ obtained by merging a triangle and a tree at one vertex. For $|V(H)| = n$, their theorem asserts, among other cases, the lower formula $\mathrm{uB}(H) \geq (n+2)(n-3)$ for $n \geq 7$. The following small graph disproves that assertion at $n = 7$.

Theorem 3.1. *There exists a graph $G \in \mathcal{U}_{7,3}$ with $\mathrm{uB}(G) = 34$. Hence the inequality*

$$\mathrm{uB}(G) \geq (n+2)(n-3)$$

for all merged graphs $C_3 \cdot T$ of order $n \geq 7$ fails at $n = 7$.

Proof. Let

$$V(G) = \{r, a, b, x, y, p, q\}, \qquad E(G) = \{ra, rb, ab, rx, xy, yp, yq\}.$$

Thus r, a, b span a triangle; a path rxy is attached at r; and two leaves p, q are attached at y. The construction is shown in Figure 1. For an unordered pair put

$$\Delta(u,v) = |n_G(u,v) - n_G(v,u)|.$$

A direct calculation gives Table 1. Summing the last column yields

$$\mathrm{uB}(G) = 8 + 1 + 2 + 1 + 6 + 10 + 4 + 2 = 34.$$

For $n = 7$, the asserted lower bound equals $(7+2)(7-3) = 36$, so the inequality is false. Finally, the graph is indeed a merged graph $C_3 \cdot T$: take the tree on vertices $\{r, x, y, p, q\}$ with edges rx, xy, yp, yq and identify its root r with a vertex of C_3. $\square$

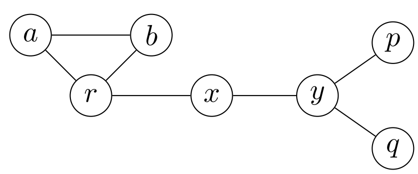

Figure 1: The seven-vertex counterexample: a triangle with a pendant path rxy and two leaves at y.

Remark 3.2. *The graph in Theorem 3.1 is not merely a girth-three unicyclic graph; it lies in the merged class $C_3 \cdot T$ considered in [4]. Thus any formula that gives 36 as the minimum for this merged class at order 7 needs an additional correction.*

Table 1: Pair contributions for the graph in Theorem 3.1.

Vertex-pair type	Number of pairs	Contribution per pair	Total
ra, rb	2	4	8
ab	1	0	0
rx	1	1	1
ry	1	0	0
rp, rq	2	1	2
xy	1	1	1
xp, xq	2	3	6
yp, yq	2	5	10
pq	1	0	0
xa, xb	2	2	4
ya, yb	2	1	2
pa, pb, qa, qb	4	0	0
Total	21		34

4 Exact minimum among all unicyclic graphs and endpoint values for fixed cycle length

Before fixing the cycle length, the global minimum over all unicyclic graphs is immediate.

Proposition 4.1. *Among all unicyclic graphs of order $n \geq 3$, the minimum value of distance-unbalancedness is 0. Equality holds if and only if the graph is the cycle C_n.*

Proof. The cycle C_n has, for every pair of vertices u, v, a reflection automorphism interchanging u and v. Hence the map induced by this reflection preserves the remaining vertex set and interchanges the two inequalities defining $n_{C_n}(u, v)$ and $n_{C_n}(v, u)$. Thus $\mathrm{uB}(C_n) = 0$.

Conversely, let G be a unicyclic graph that is not a cycle. Then G has a leaf ℓ. Let s be the unique neighbor of ℓ. For every vertex $w \neq \ell$, every path from ℓ to w starts with the edge ℓs, and so

$$\mathrm{dist}_G(s, w) < \mathrm{dist}_G(\ell, w).$$

Therefore $n_G(\ell, s) = 1$ and $n_G(s, \ell) = n - 1$. The pair $\{\ell, s\}$ contributes $n - 2 > 0$ to $\mathrm{uB}(G)$, and hence $\mathrm{uB}(G) > 0$. □

Now fix the length c of the unique cycle.

Proposition 4.2. *Let $c \geq 3$.*

(i) *If $n = c$, then $\mathcal{U}_{c,c} = \{C_c\}$ and*

$$m_c(c) = M_c(c) = 0.$$

(ii) *If $n = c+1$, then every graph in $\mathcal{U}_{c+1,c}$ is isomorphic to the graph obtained from C_c by adding one leaf at a cycle vertex, and*

$$m_c(c+1) = M_c(c+1) = \begin{cases} \dfrac{c^2 + 4c - 9}{2}, & c \text{ odd}, \\ \dfrac{c^2 + 4c - 10}{2}, & c \text{ even}. \end{cases}$$

Proof. Part (i) is immediate. For (ii), let L_{c+1} be the graph obtained from C_c by adding a leaf x adjacent to a root vertex r. Since C_c is vertex-transitive, this graph is unique up to isomorphism.

It remains to compute $\mathrm{uB}(L_{c+1})$. For pairs of two cycle vertices, the cycle vertices themselves remain balanced; only the added leaf x can break the balance. Thus the contribution over all cycle-cycle pairs equals the number of unordered pairs of cycle vertices that have different distances from r in C_c.

Write first $c = 2q + 1$. When the root r is omitted and one walks once around the cycle, the distances from r are

$$1, 2, \ldots, q, q, \ldots, 2, 1.$$

Equivalently, the distance layers from r have sizes $1, 2, 2, \ldots, 2$. Hence the cycle-cycle contribution is

$$\binom{c}{2} - q = 2q^2,$$

because the only cycle-cycle pairs not separated by the new leaf are the q pairs lying in the same nonzero distance layer. For the leaf-cycle pairs, the pair $\{x, r\}$ contributes $2q$, the two cycle vertices at distance 1 from r contribute q each, and every remaining cycle vertex contributes 1. Thus these c pairs contribute

$$2q + 2q + 2(q - 1) = 6q - 2.$$

Consequently

$$\mathrm{uB}(L_{c+1}) = 2q^2 + 6q - 2 = \frac{c^2 + 4c - 9}{2}.$$

Now write $c = 2q$. When the root r is omitted and one walks once around the cycle, the distances from r are

$$1, 2, \ldots, q, \ldots, 2, 1.$$

Equivalently, the distance layers from r have sizes $1, 2, \ldots, 2, 1$. Hence the cycle-cycle contribution is

$$\binom{c}{2} - (q - 1) = 2q^2 - 2q + 1.$$

For the leaf-cycle pairs, the pair $\{x, r\}$ contributes $2q - 1$, the two vertices at distance 1 from r contribute $q - 1$ each, and every remaining cycle vertex contributes 1. Thus the leaf-cycle contribution is

$$(2q - 1) + 2(q - 1) + (2q - 3) = 6q - 6.$$

Therefore

$$\mathrm{uB}(L_{c+1}) = 2q^2 + 4q - 5 = \frac{c^2 + 4c - 10}{2}.$$

□

5 Minimum for fixed cycle length: test families and conjectures

The exact finite-order value of $m_c(n)$ for general c appears to involve several exceptional small cases. This section records explicit low-unbalanced families and the conjectural asymptotic picture suggested by exhaustive enumeration. Let

$$t = n - c$$

be the number of vertices outside the cycle. We write $O_c(\cdot)$ for Landau notation in which the implicit constant may depend on the fixed cycle length c.

5.1 Bouquets and moving leaves

A useful family is obtained by moving pendant leaves among selected vertices of the fixed cycle. Let $B_{c,t}(d;a,b)$ denote the graph obtained from C_c by choosing two cycle vertices A and B at cycle-distance d and adding a leaves to A and b leaves to B, where $a+b=t$. The cases $a=0$ or $b=0$ are allowed. For $c=3$ we also use $S_{3,t}$ for the graph obtained from C_3 by adding all t leaves at one cycle vertex.

The following proposition gives explicit values for these test families. These formulae are not asserted to be the exact minima for all small t; rather, they provide constructive upper bounds and the leading constants in the conjectures below.

Proposition 5.1. *Let $t=n-c$.*

(a) *If $S_{3,t}$ is obtained from C_3 by adding all t leaves at one cycle vertex, then*

$$\mathrm{uB}(S_{3,t}) = t^2+5t.$$

(b) *Let $c\geq 4$ be even, and put the two support vertices at distance $c/2-1$ on C_c. If $a=\lfloor t/2\rfloor$ and $b=\lceil t/2\rceil$, then*

$$\mathrm{uB}\big(B_{c,t}(c/2-1;a,b)\big) = \begin{cases} \dfrac{3}{2}t^2+(5c-11)t, & t \text{ even},\\ \dfrac{7}{4}t^2+(5c-11)t+\dfrac{2c^2-12c+17}{4}, & t \text{ odd}.\end{cases}$$

(c) *Let $c\geq 5$ be odd, and put the two support vertices at distance $(c-1)/2$ on C_c. If $a=\lfloor t/2\rfloor$ and $b=\lceil t/2\rceil$, then*

$$\mathrm{uB}\big(B_{c,t}((c-1)/2;a,b)\big) = \begin{cases} 2t^2+\dfrac{9c-19}{2}t, & t \text{ even},\\ \dfrac{9}{4}t^2+\dfrac{9c-19}{2}t+\dfrac{2c^2-10c+11}{4}, & t \text{ odd}.\end{cases}$$

Proof. We give the finite-type count explicitly. Put the support vertices at $A=0$ and $B=d$ on the labelled cycle C_c with vertex set $[c]$. Let L_A and L_B be the sets of leaves attached to A and B, and put $|L_A|=a$, $|L_B|=b$. Leaves attached to the same support are twins, so any pair of leaves in the same L_A or the same L_B contributes 0.

We use the signed imbalance

$$D_G(u,v) = n_G(u,v) - n_G(v,u),$$

so that the contribution of $\{u,v\}$ is $|D_G(u,v)|$. All contributions are split into three classes: cycle-cycle pairs, cycle-leaf pairs, and leaf-leaf pairs with different supports.

Cycle-cycle pairs. For two cycle vertices i,j, the c cycle vertices give zero signed contribution, because C_c has a reflection interchanging i and j. A leaf at a support vertex $S\in\{A,B\}$ is closer to i than to j exactly when S is closer to i than to j on the cycle. Hence

$$\Sigma_{CC} = \sum_{0\leq i<j<c} \left| a\varepsilon\big(\rho_c(A,i),\rho_c(A,j)\big) + b\varepsilon\big(\rho_c(B,i),\rho_c(B,j)\big)\right|. \tag{2}$$

Cycle-leaf pairs. Let x_S be a leaf attached to $S\in\{A,B\}$, and let i be a cycle vertex. Let R denote the other support vertex. The signed contribution of x_S itself is 1. A cycle vertex z contributes

$$\varepsilon\big(\rho_c(z,S)+1,\rho_c(z,i)\big).$$

Another leaf at the same support S contributes

$$\varepsilon\big(2, 1+\rho_c(S,i)\big),$$

and a leaf at the other support R contributes

$$\varepsilon\big(2+\rho_c(R,S), 1+\rho_c(R,i)\big).$$

Therefore, if $s_A = a$ and $s_B = b$, then

$$\begin{aligned} D_G(x_S, i) = 1 + \sum_{z=0}^{c-1} \varepsilon\big(\rho_c(z,S)+1, \rho_c(z,i)\big) \\ + (s_S - 1)\varepsilon\big(2, 1+\rho_c(S,i)\big) + s_R \varepsilon\big(2+\rho_c(R,S), 1+\rho_c(R,i)\big), \end{aligned} \tag{3}$$

and the total cycle-leaf contribution is

$$\Sigma_{CL} = a \sum_{i=0}^{c-1} |D_G(x_A, i)| + b \sum_{i=0}^{c-1} |D_G(x_B, i)|. \tag{4}$$

Leaves at opposite supports. For $x_A \in L_A$ and $x_B \in L_B$, the cycle vertices contribute zero in total by the reflection of C_c interchanging A and B. The leaves themselves contribute $a-b$ to the signed imbalance $D_G(x_A, x_B)$. Thus each such pair contributes $|a-b|$, and

$$\Sigma_{LL} = ab|a-b|. \tag{5}$$

Evaluating the explicit sums (2)–(5) in the required cases gives Table 2. The table separates the same three classes of pairs, so each entry has a direct interpretation. For example, in the row $c = 2q$, $t = 2s$, we have $a = b = s$ and $d = q-1$. In (2), the quantity

$$\big|\varepsilon\big(\rho_c(A,i), \rho_c(A,j)\big) + \varepsilon\big(\rho_c(B,i), \rho_c(B,j)\big)\big|$$

takes value 2 for q cycle pairs, value 1 for $2q-2$ cycle pairs, and value 0 for the remaining cycle pairs. This gives

$$\Sigma_{CC} = s\{2q + (2q-2)\} = (4q-2)s.$$

For one fixed leaf at A, the values of $|D_G(x_A, i)|$ as i runs around the cycle are

$$2q+2s-2, \quad q+s-1, \underbrace{2, \ldots, 2}_{2q-3 \text{ times}}, q-1,$$

whose sum is $3s + 8q - 10$. The same sum holds for one fixed leaf at B, so

$$\Sigma_{CL} = 2s(3s + 8q - 10) = 6s^2 + (16q-20)s.$$

Since $a = b$, equation (5) gives $\Sigma_{LL} = 0$. This verifies the whole row $c = 2q, t = 2s$. The other rows are obtained by the same displayed formulae; the terms with $s(s+1)$ occur exactly when $|L_B| = |L_A| + 1$, because then each opposite-support leaf pair has imbalance 1.

Adding the three columns in each row gives

$$\begin{aligned} S_{3,t}: &\quad t^2 + 5t, \\ c = 2q,\ t = 2s: &\quad 6s^2 + (20q-22)s, \\ c = 2q,\ t = 2s+1: &\quad 7s^2 + (20q-15)s + 2q^2 + 4q - 5, \\ c = 2q+1,\ t = 2s: &\quad 8s^2 + (18q-10)s, \\ c = 2q+1,\ t = 2s+1: &\quad 9s^2 + (18q-1)s + 2q^2 + 6q - 2. \end{aligned}$$

Rewriting these expressions in terms of c and t gives exactly the formulae displayed in the statement.

□

Table 2: Finite-type contribution sums for Proposition 5.1. Here q and s are defined by the parity of c and t in the first column. The column "cycle-leaf" includes all pairs consisting of one leaf and one cycle vertex, including support vertices.

Case	cycle-cycle	cycle-leaf	opposite-support leaves
$S_{3,t}$	$2t$	$t(t+1)+2t$	0
$c=2q,\ t=2s$	$(4q-2)s$	$6s^2+(16q-20)s$	0
$c=2q,\ t=2s+1$	$2q^2-2q+1+(4q-2)s$	$6s^2+(16q-14)s+6q-6$	$s(s+1)$
$c=2q+1,\ t=2s$	$2qs$	$8s^2+(16q-10)s$	0
$c=2q+1,\ t=2s+1$	$2q^2+2qs$	$8s^2+(16q-2)s+6q-2$	$s(s+1)$

5.2 Computational evidence for the minimum

Table 3 gives exact values obtained by exhaustive enumeration of rooted-cycle representations. The enumeration method is described in Section 7. The exceptional behavior at small t is visible; for instance, the counterexample in Theorem 3.1 is responsible for $m_3(7)=34$ rather than 36.

Table 3: Exact values of $m_c(c+t)$ for $0 \le t \le 8$ and $3 \le c \le 6$.

$t=n-c$	$m_3(3+t)$	$m_4(4+t)$	$m_5(5+t)$	$m_6(6+t)$
0	0	0	0	0
1	6	11	18	25
2	12	20	34	36
3	20	39	60	71
4	34	56	78	88
5	50	79	106	135
6	66	108	138	156
7	84	141	180	217
8	104	168	228	240

The data and Proposition 5.1 suggest the following asymptotic form.

Conjecture 5.2. *Let $c \ge 3$ be fixed and let $t=n-c\to\infty$.*

(i) *For $c=3$,*

$$m_3(3+t)=t^2+O(t),$$

and more precisely $m_3(3+t)=t^2+5t$ for all sufficiently large t, with the exceptional value $m_3(7)=34$.

(ii) *For even $c \ge 4$,*

$$m_c(c+t)=\frac{3}{2}t^2+O_c(t)$$

when t is even. When t is odd, the best construction may require a bounded local correction; the expected value is still quadratic with leading coefficient between $3/2$ and $7/4$, and the uncorrected two-bouquet family has leading coefficient $7/4$.

(iii) *For odd $c \ge 5$,*

$$m_c(c+t)=2t^2+O_c(t).$$

Remark 5.3. *Conjecture 5.2 is deliberately stated as an asymptotic conjecture. The small cases in Table 3 show that short subdivided stars, brooms, and other local corrections can beat the simplest bouquet constructions when t is small. Any exact finite-order theorem should account for these exceptions.*

6 Maximum for fixed cycle length

In contrast with the minimum, the maximum has a clean leading term for every fixed cycle length.

Theorem 6.1. *For every fixed integer $c \geq 3$,*

$$M_c(n) = \frac{1}{2}n^3 + o(n^3) \qquad \text{as } n \to \infty.$$

Proof. For every graph G of order n and every unordered pair $\{u, v\}$, the vertices u and v themselves are counted on opposite sides, and hence

$$|n_G(u, v) - n_G(v, u)| \leq n - 2.$$

Therefore

$$M_c(n) \leq \binom{n}{2}(n-2) = \frac{1}{2}n^3 + O(n^2).$$

For the lower bound, put $m = n - c + 1$. Kramer and Rautenbach [3] proved that the maximum distance-unbalancedness among trees of order m is

$$\frac{1}{2}m^3 + o(m^3).$$

Choose a tree T_m of order m with

$$\mathrm{uB}(T_m) = \frac{1}{2}m^3 + o(m^3),$$

select a root vertex $r \in V(T_m)$, and identify r with one vertex of a new cycle C_c. Equivalently, add $c - 1$ new vertices and the c cycle edges through r. The resulting graph G lies in $\mathcal{U}_{n,c}$.

Distances between old vertices of T_m are unchanged: any path leaving T_m must leave and re-enter through the same vertex r, so it only adds a positive cycle detour. Fix an old pair $\{u, v\} \subseteq V(T_m)$. The old vertices contribute the same signed imbalance in G as in T_m. The $c - 1$ new cycle vertices can change this signed imbalance by at most $c - 1$ in absolute value. Hence, by the reverse triangle inequality,

$$|n_G(u, v) - n_G(v, u)| \geq |n_{T_m}(u, v) - n_{T_m}(v, u)| - (c - 1).$$

Summing over all unordered pairs of old vertices gives

$$\mathrm{uB}(G) \geq \mathrm{uB}(T_m) - (c-1)\binom{m}{2} = \frac{1}{2}m^3 + o(m^3) - O_c(m^2).$$

Since $m = n - c + 1 = n + O_c(1)$, this is

$$\mathrm{uB}(G) \geq \frac{1}{2}n^3 + o(n^3).$$

Together with the upper bound, this proves the theorem. □

Remark 6.2. *The sign in the last lower bound is necessarily a minus sign. For a fixed old pair $\{u, v\}$, the new cycle vertices may increase or decrease the old signed imbalance, depending on which of u and v is closer to the attachment vertex r. The asymptotic result only requires the uniform error bound $c - 1$ per old pair.*

The exact finite maximum seems to inherit the integer optimization issues from the tree case. Table 4 gives exact values for small $t = n - c$.

Table 4: Exact values of $M_c(c + t)$ for $0 \le t \le 8$ and $3 \le c \le 6$.

$t = n - c$	$M_3(3 + t)$	$M_4(4 + t)$	$M_5(5 + t)$	$M_6(6 + t)$
0	0	0	0	0
1	6	11	18	25
2	16	24	40	57
3	32	47	72	97
4	57	80	118	154
5	92	126	176	232
6	140	185	251	324
7	203	263	346	439
8	282	362	464	581

Conjecture 6.3. *For every fixed c, the finite extremal graphs for $M_c(n)$ are obtained from subdivided-star-like extremal trees by adding a C_c at a central or near-central vertex, with only bounded local modifications depending on congruence and small-order effects.*

7 Enumeration method

This section describes the exhaustive enumeration used for Tables 3 and 4. It is included to make the computational claims reproducible.

A rooted tree is encoded recursively by the sorted multiset of encodings of the rooted branches at its root. The one-vertex rooted tree is encoded by the empty tuple (). If the root has rooted branches with encodings $\alpha_1, \dots, \alpha_k$, then the encoding of the whole rooted tree is the sorted tuple

$$(\alpha_1, \dots, \alpha_k).$$

The size of an encoding is defined recursively by

$$|()| = 1, \qquad |(\alpha_1, \dots, \alpha_k)| = 1 + \sum_{i=1}^{k} |\alpha_i|.$$

Let $\mathcal{R}_s$ be the set of all encodings of rooted trees with s vertices. Then $\mathcal{R}_1 = \{()\}$, and for $s \ge 2$ the set $\mathcal{R}_s$ is generated by taking all sorted multisets of elements from $\bigcup_{j=1}^{s-1} \mathcal{R}_j$ whose total size is $s - 1$. This is the standard canonical generation of unlabeled rooted trees; sorting the branch encodings removes the dependence on the order of branches.

For fixed c and t, every graph in $\mathcal{U}_{c+t,c}$ is represented by a c-tuple

$$(T_0, T_1, \dots, T_{c-1})$$

of rooted trees satisfying

$$\sum_{i=0}^{c-1} |V(T_i)| = c + t, \qquad \text{equivalently} \qquad \sum_{i=0}^{c-1} (|V(T_i)| - 1) = t.$$

The exhaustive enumeration proceeds as follows.

1. Generate the rooted-tree lists $\mathcal{R}_s$ for $1 \le s \le t + 1$.

2. Generate every weak composition $(e_0, \dots, e_{c-1})$ of t, where $e_i = |V(T_i)| - 1$ is the number of non-cycle vertices in the tree attached at the ith cycle vertex.

3. For this composition, choose each T_i independently from $\mathcal{R}_{e_i+1}$.

4. Optionally quotient by the dihedral symmetries of the cycle by replacing the tuple code with the lexicographically least code among all rotations and reversals. This step gives one representative per isomorphism class. It is not needed for computing only minima and maxima, because duplicate tuples cannot change an extremal value.

5. Build the graph $C_c(T_0, \dots, T_{c-1})$. Compute all-pairs distances, either by breadth-first search from each vertex or by using (1) together with within-tree distances. Finally evaluate

$$\mathrm{uB}(G) = \sum_{\{u,v\} \in \binom{V(G)}{2}} |n_G(u, v) - n_G(v, u)|.$$

For the ranges in Tables 3 and 4, this enumeration is small. The rooted-tree counts needed up to size 9 are

$$1, 1, 2, 4, 9, 20, 48, 115, 286,$$

which correspond to sizes $1, 2, \dots, 9$. As an independent check on the arithmetic, the values in Tables 3 and 4 were evaluated from the definition of uB using all-pairs breadth-first-search distances.

8 Concluding remarks

The prescribed-cycle-length problem has two different levels of difficulty. The maximum is asymptotically governed by the corresponding tree problem, and Theorem 6.1 gives the sharp cubic leading term. The minimum is more delicate. The global unicyclic minimum is trivial because cycles are distance-balanced in the complete-pair sense, but the fixed-c minimum with $n > c$ is positive and sensitive to the parity of c, the parity of $n - c$, and small local corrections.

The seven-vertex counterexample in Theorem 3.1 shows that even the triangle case has exceptional behavior. A natural next step is to prove Conjecture 5.2, beginning with an exact corrected theorem for $c = 3$ and then an asymptotic lower bound for the two-bouquet structures suggested by even and odd cycles.

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